

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: 4.2 PROBABILITY I

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use what you learned about Mama Njeri's mandazi stall at Gikomba Market to answer the driving question: How can we use mathematics to measure and predict the likelihood of uncertain everyday events, so we can make smarter decisions even when we cannot be certain of the outcome? Write your answers in the space under each prompt. Show your working for every calculation and write probabilities as fractions or decimals between 0 and 1. Use words from the unit, such as random experiment, sample space, event, relative frequency, mutually exclusive and independent. Always link your answer back to Mama Njeri's stall.

Part 1: Why does the fraction of good days settle down?

Mama Njeri kept a 60-day notebook. A 'good day' is a day when she sells at least 80% of her stock. Suppose 24 of the 60 days were good days. (a) Calculate the experimental probability of a good day. (b) In your own words, explain what this number tells Mama Njeri and what it does NOT tell her about tomorrow. (c) Your class plotted running relative frequencies for coins and bottle caps. Use that evidence to explain why the fraction of good days in her notebook becomes steadier the longer she keeps records, even though each single day is unpredictable.

(a) Experimental probability $P(E) = \text{number of times the event occurs} \div \text{total number of trials}$. $P(\text{good day}) = 24 \div 60 = 0.4$. (b) It means that over many days of trading, about 40% of her days (4 days in every 10) will be good days. It does NOT mean tomorrow is guaranteed to be good or bad. Each day is still a random experiment with an uncertain outcome. (c) In our coin and bottle-cap graphs the relative frequency jumped around when there were few trials, but as we pooled more trials each line settled near its own stable value (the coin near 0.5). The same happens with Mama Njeri's days. With few days, chance variation has a big effect. With more days, the ups and downs average out and the relative frequency moves closer to the true probability of a good day. This is why a longer notebook is more useful for planning, even though no single day can be predicted exactly.

Part 2: Reasoning before the stall opens

(a) List the sample space when

(a) Sample space = {HH, HT, TH, TT}, which has 4 equally likely outcomes. (b) Exactly one head happens in HT and TH, so there

two fair coins are tossed together. (b) Use your sample space to find the theoretical probability of getting exactly one head. (c) Explain how theoretical probability helps us understand why experimental relative frequencies settle down over many trials, and why Mama Njeri can reason about probability even before she opens her stall.	are 2 favourable outcomes. Theoretical $P(E) = \text{favourable outcomes} \div \text{total outcomes in the sample space} = 2 \div 4 = 1/2 = 0.5$. (c) The theoretical probability is the true fraction of favourable outcomes in the sample space. When we do many trials, the experimental relative frequency gets closer to this fraction. For a fair coin the theoretical value 0.5 matched the value our graph was settling towards. Mama Njeri's sales also follow an underlying probability space, so there is a true probability of a good day. This is why she sees steady long-run patterns, and why she can think about likelihoods before opening her stall, even though any single day remains uncertain.
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Part 3: Events that overlap

Over 20 days in Nakuru, a seller recorded days with HIGH SALES (H) and days with HEAVY RAIN (R). There were 9 days with high sales and 6 days with heavy rain. On 3 of the days there was both high sales and heavy rain. (a) How many of the 20 days had high sales OR heavy rain (or both)? Use a Venn diagram to help you. (b) Write your answer to (a) as a probability $P(H \cup R)$. (c) A student finds the number of days by working out $9 + 6$. Is this correct? Explain, and write the Addition Law of Probability for any two events A and B. (d) Explain how the Addition Law changes when two events are mutually exclusive. Give one example from Mama Njeri's stall.	(a) Days in the union $= n(H) + n(R) - n(H \cap R) = 9 + 6 - 3 = 12$ days. In the Venn diagram: 6 days high sales only, 3 days both, 3 days heavy rain only, and 8 days neither; $6 + 3 + 3 = 12$. (b) $P(H \cup R) = 12 \div 20 = 0.6$, or 60%. (c) No. $9 + 6 = 15$ counts the 3 days with both events twice: once in H and once in R. In a Venn diagram these days sit in the overlap, which belongs to both circles. We must subtract the overlap once. Addition Law: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. (d) Mutually exclusive events cannot both happen in the same trial, so $A \cap B$ is empty and $P(A \cap B) = 0$. The law becomes $P(A \cup B) = P(A) + P(B)$. Example: 'Mama Njeri sells out' and 'half her mandazi are unsold' cannot happen on the same day, so their probabilities can be added directly.
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Part 4: One day after another

Suppose the probability that any day is a good day for Mama Njeri is 0.4, and each day is	(a) Day 1: Good (G) 0.4, Bad (B) 0.6. From each of these, Day 2: G 0.4, B 0.6. Paths: $GG = 0.4 \times 0.4 = 0.16$; $GB = 0.4 \times 0.6 = 0.24$; $BG = 0.6 \times 0.4 = 0.24$; $BB = 0.6 \times 0.6 = 0.36$. We multiply along the branches. (b) $P(\text{two good days}) = 0.16$. $P(\text{two bad days}) = 0.36$. Exactly one good day happens through two paths, GB or BG, so we add across paths: $0.24 + 0.24 = 0.48$. (c) The four
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independent of the others. (a) Draw a two-day probability tree diagram. Label every branch and every path with its probability. (b) Find the probability of two good days in a row, the probability of two bad days in a row, and the probability of exactly one good day in the two days. (c) Explain how you could check that your path probabilities are all correct. (d) A classmate says that independent events and mutually exclusive events mean the same thing. Do you agree? Explain the difference using an example from Mama Njeri's stall.	paths cover every possible outcome of the two days, so their probabilities must add to 1: $0.16 + 0.24 + 0.24 + 0.36 = 1$. (d) No, I disagree. Mutually exclusive events cannot both happen in the same trial, so $P(A \cap B) = 0$. Independent events can both happen, and one does not change the probability of the other, so $P(A \cap B) = P(A) \times P(B)$. Example: 'a day is good' and 'a day is bad' are mutually exclusive because they cannot both happen on one day. But 'yesterday was good' and 'today is good' are independent: both can happen, and yesterday's result does not change today's probability. So the two ideas are different.
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Part 5: Advising Mama Njeri and answering the driving question

Write a short recommendation to Mama Njeri about how to plan her daily batch of mandazi. (a) Use at least three tools from this unit (for example experimental probability, the Addition Law, the Multiplication Law or a tree diagram), and use numbers from your earlier answers. (b) Explain why your advice is a better strategy than guessing, even though probability cannot tell her what will happen tomorrow. (c) Using your recommendation, answer the driving question in two or three sentences.	(a) Dear Mama Njeri, from your notebook, about 40% of your days are good days ($P = 24 \div 60 = 0.4$), so about 60% of your days are not good days and you should not fry a huge batch every morning. Because each day is independent, yesterday's sales do not change today's chances. Our two-day tree shows that two good days in a row has probability 0.16, two bad days in a row has probability 0.36, and exactly one good day has probability 0.48, which is the most likely result. So plan your stock for a mix of good and bad days, not for good days every day. On days when heavy rain could lower foot traffic, remember that events overlap: use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ so you do not overcount, as in the Nakuru data where 12 out of 20 days (0.6) had high sales or heavy rain. Choose a moderate batch size that matches these probabilities and keep updating your notebook. (b) Guessing ignores the pattern in your records. Using relative frequency, you plan for what happens in the long run, so over many days you lose less money from unsold mandazi and from turned-away customers. We still cannot say what tomorrow will be, because each day is a random experiment. (c) Mathematics lets us measure how likely uncertain events are, using experimental probability from records and theoretical probability from sample spaces. We can combine events using the Addition and Multiplication Laws and tree diagrams. Even though individual outcomes are uncertain, patterns over many trials are predictable, so we can make smarter decisions without being certain.
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RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Calculating and interpreting probabilities	All probabilities are calculated correctly from the data or sample space, shown with working, and interpreted as long-run likelihoods that do not guarantee a single day.	Calculations are mostly correct with some working; interpretation is mostly correct but may imply certainty for one day.	Calculations contain errors or lack working; interpretation of probability is unclear or treats it as a prediction of one outcome.
Explaining stabilisation of relative frequency	Clearly explains, using class coin and bottle-cap evidence, why more trials make relative frequency settle near the true probability while single days stay unpredictable.	States that more trials give a steadier fraction and refers to evidence, but the reason is only partly explained.	Does not explain why the fraction settles, or does not use evidence from the experiments.
Using the Addition and Multiplication Laws and tree diagrams	Chooses and applies the correct law, explains double counting, builds a fully labelled tree diagram, and clearly distinguishes independent from mutually exclusive events.	Applies the laws and tree diagram mostly correctly; the explanation of double counting or of independent versus mutually exclusive is incomplete.	Uses the wrong law, has an incomplete or unlabelled tree diagram, or confuses independent and mutually exclusive events.
Using probability to make a decision	Gives a justified recommendation to Mama Njeri that uses several tools and specific numbers, and answers the driving question including the role of uncertainty.	Gives a reasonable recommendation using some tools and numbers, and a partly complete answer to the driving question.	Recommendation is vague or not supported by probability evidence, and the driving question is not answered.
Mathematical communication	Uses unit vocabulary accurately and writes in clear, organised sentences.	Uses most vocabulary correctly; writing is generally clear.	Vocabulary is missing or misused; writing is hard to follow.

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